

Facebook Data Psychometric Analysis Guide

End-to-End Workflow: Downloading, Parsing, Sentiment Analysis, and Visualization on Debian / Parrot OS

1. Overview & Setup

This manual provides complete step-by-step instructions to export your Facebook data archive and execute psychological text analysis using Python and NLP (Natural Language Processing) tools on Debian or Parrot OS.

Prerequisites & System Packages

Open your terminal and ensure Python 3, `pip`, and core development tools are installed:

```
sudo apt update
sudo apt install -y python3 python3-pip python3-pandas python3-nltk python3-matplotlib
```

2. Exporting Your Facebook Archive

1. Log into Facebook and navigate to **Settings & Privacy > Settings**.
2. In the left sidebar / Accounts Center, select **Your Information and Permissions > Download your information**.
3. Click **Download or transfer information** and select **Specific types of information**.
4. Check **Posts**, **Comments**, and **Messages** (or select all data).
5. **Crucial Step:** Set the format to **JSON** (do not choose HTML). Select **All Time** for date range.
6. Submit the request and wait for the download link via notification/email.

Why JSON format?

JSON provides raw structured data that Python functions like `json.load()` can parse natively. HTML is designed for web display and requires complex web scraping.

3. Directory Structure & Character Encoding Quirk

Once downloaded, extract the ZIP file to your workspace directory (e.g., `~/Downloads/facebook-archive/`). Your main post file will be located at:

```
your_facebook_activity/posts/your_posts_1.json
```

Fixing Facebook's Character Encoding

Facebook JSON files encode non-ASCII characters (emojis, apostrophes) as Latin-1 byte strings. Use this helper function in Python to fix it:

```
def fix_fb_encoding(text):
    if isinstance(text, str):
        try:
            return text.encode('latin1').decode('utf-8')
        except (UnicodeEncodeError, UnicodeDecodeError):
            return text
    return text
```

4. Complete Python Script (`analyze\_fb.py`)

Create a script named `analyze_fb.py` inside your extracted Facebook root folder. This script performs sentiment analysis using VADER, calculates word frequencies, and generates a time-series plot of your sentiment history.

```

import json
import os
import re
from collections import Counter
import pandas as pd
import nltk
from nltk.sentiment.vader import SentimentIntensityAnalyzer
from nltk.corpus import stopwords
import matplotlib.pyplot as plt

# Download required NLTK resources
nltk.download('vader_lexicon', quiet=True)
nltk.download('stopwords', quiet=True)

# 1. Locate and Load the Posts JSON
json_path = 'your_facebook_activity/posts/your_posts_1.json'

if not os.path.exists(json_path):
    print(f"Error: Could not find '{json_path}'. Run from unzipped Facebook folder.")
    exit(1)

with open(json_path, 'r', encoding='utf-8') as f:
    data = json.load(f)

def fix_fb_encoding(text):
    if isinstance(text, str):
        try:
            return text.encode('latin1').decode('utf-8')
        except (UnicodeEncodeError, UnicodeDecodeError):
            return text
    return text

# 2. Extract Text and Timestamps
posts = []
for entry in data:
    if "data" in entry and entry["data"]:
        text = entry["data"][0].get("post", "")
        time = entry.get("timestamp", None)
        if text:
            posts.append({
                "timestamp": pd.to_datetime(time, unit='s'),
                "text": fix_fb_encoding(text)
            })

df = pd.DataFrame(posts)
print(f"Successfully loaded {len(df)} Facebook posts!")
")

if df.empty:
    print("No textual status updates found.")
    exit(0)

# Sort chronologically
df = df.sort_values('timestamp').reset_index(drop=True)

# 3. Sentiment Analysis (VADER)
sia = SentimentIntensityAnalyzer()
df['sentiment'] = df['text'].apply(lambda x: sia.polarity_scores(x)['compound'])

```

```

print("--- OVERALL SENTIMENT SUMMARY ---")
print(f"Average Sentiment Score: {df['sentiment'].mean():.3f} (-1.0 to +1.0 scale)")
print(f"Most Positive Post Score: {df['sentiment'].max():.3f}")
print(f"Most Negative Post Score: {df['sentiment'].min():.3f}")
")

# 4. Word Frequency Analysis
stop_words = set(stopwords.words('english'))
custom_stops = {'https', 'http', 'com', 'like', 'get', 'one', 'would'}
stop_words.update(custom_stops)

all_words = []
for text in df['text']:
    cleaned = re.sub(r'^a-zA-Z\s', '', text).lower()
    words = [w for w in cleaned.split() if w not in stop_words and len(w) > 2]
    all_words.extend(words)

word_counts = Counter(all_words)

print("--- TOP 20 MOST FREQUENT WORDS ---")
for word, count in word_counts.most_common(20):
    print(f"{word:<15}: {count}")

# 5. Save Detailed CSV Results
df.to_csv("fb_sentiment_results.csv", index=False)
print("
Saved processed data to 'fb_sentiment_results.csv'")

# 6. Generate Time-Series Visualization
df.set_index('timestamp', inplace=True)
monthly_sentiment = df['sentiment'].resample('ME').mean()

plt.figure(figsize=(12, 6))
plt.plot(monthly_sentiment.index, monthly_sentiment.values, color='#2b6cb0', linewidth=2,
marker='o', fontsize=10)
plt.axhline(0, color='#e53e3e', linestyle='--', linewidth=1, label='Neutral Sentiment (0.0)')

plt.title('Facebook Sentiment Trajectory Over Time (Monthly Average)', fontsize=14,
fontweight='bold', pad=15)
plt.xlabel('Year', fontsize=11)
plt.ylabel('VADER Compound Sentiment Score (-1 to +1)', fontsize=11)
plt.grid(True, linestyle=':', alpha=0.6)
plt.legend()

plt.tight_layout()
plot_filename = "facebook_sentiment_trend.png"
plt.savefig(plot_filename, dpi=300)
print(f"Saved visualization plot to '{plot_filename}'")

```

5. Running the Analysis & Interpreting Output

Execute the script from your terminal inside the unzipped archive directory:

```
python3 analyze_fb.py
```

Generated Deliverables

File Name	Format	Description
fb_sentiment_results.csv	CSV Spreadsheet	Contains every individual post with timestamp, full text, and VADER sentiment compound score (-1 to +1).
facebook_sentiment_trend.png	PNG Image (300 DPI)	High-resolution graph plotting monthly emotional trajectories over time.

Generated for Debian / Parrot OS Psychometric Analysis Workflows.